

Q37. Wi-Fi Security for IoT

Research and write 250 words on: Common Wi-Fi attack vectors on IoT devices (WPA2 vulnerabilities, evil twin attacks), Best practices for securing ESP32 Wi-Fi connections (certificate pinning, WPA3), and Why hardcoding passwords in IoT source code is dangerous. Reference at least 2 sources in your answer.

Ans. IoT devices like the ESP32 are essentially mini-computers that are always connected to the internet. Because they are often small and cheap, they are favourite targets for hackers.

1. Common Wi-Fi Attack Vectors

WPA2 Vulnerabilities (KRACK):

WPA2 is the standard security for most Wi-Fi networks. A "KRACK" attack tricks the network into letting a hacker "peek" at the information being sent between your device and the router, even without the Wi-Fi password.

Evil Twin Attacks:

Imagine a hacker sets up a fake Wi-Fi network that has the exact same name as your home network. Your IoT device might accidentally connect to the "Evil Twin" instead of the real one. Once connected, the hacker can steal your data or send bad instructions to your device.

2. Best Practices for Securing ESP32

Use WPA3:

If your router supports it, use WPA3. It is the newest security standard and is specifically designed to be much harder to hack than WPA2.

Certificate Pinning:

When your device talks to a server, it usually asks, "Are you who you say you are?" The server shows a "certificate" to prove it. "Pinning" means you hardcode the server's digital "fingerprint" into your ESP32. Even if a hacker tries to trick your device with a fake certificate, the ESP32 will reject it because it doesn't match the one you pinned.

3. Why Hardcoding Passwords is Dangerous

Hardcoding (writing the password directly into your code) is a huge security "no-no":

Accidental Leaks: If you upload your code to a public site like GitHub, anyone in the world can read your password.

Physical Access: If a hacker gets hold of your device, they can extract the code from the chip. If the password is written in plain text, they now have full access to your home network.

The Fix: Use a separate configuration file that you never upload to the internet, or use secure storage settings on the chip itself.

Sources

Espressif Systems Documentation: "ESP32 Security Guidance" (Look up their official documentation for the latest security protocols).

OWASP IoT Project: Look for their "Top 10 IoT Security Vulnerabilities" guide, which explains why hardcoded credentials are a top-tier risk.